Security Engineering in Academic Distribution Systems
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1. Introduction
Modern digital distribution of academic manuscripts and educational curricula requires robust mechanisms for copyright tracing and leak attribution. When sensitive preprints or examination materials are disseminated to authorized students, watermarking systems embed unique cryptographic identifiers into distributed copies. If an unauthorized copy is subsequently leaked, investigators extract the embedded mark to determine the originating recipient.

2. Threat Model and Adversarial Capabilities
In our threat model, an authorized recipient who receives a watermarked document may attempt to sanitize the carrier to prevent tracing. The adversary may apply lossless PDF optimizations, alter catalog dictionary metadata, append extraneous commentary, crop margins, reorder pages, or rasterize the document through print-scan pipelines. A robust forensic protocol must define clear boundaries regarding which transformations permit reliable recovery.

3. Two-Column Formatting and Carrier Dynamics
Academic manuscripts commonly employ multi-column layouts using proportional serif typefaces such as Times-Roman. Two-column typography introduces narrower text frames, typically spanning 230 to 250 typographic points per column. The narrower horizontal span increases line-break frequency and alters inter-word space distribution. When evaluating covert modulation within two-column settings, parsing routines must properly track reading order across columns.

4. Mathematical Formulation of Inter-Word Kerning
In the Portable Document Format, glyph displacement within the TJ array operator is specified in thousandths of an em. By convention, negative displacement numbers translate subsequent glyphs to the right along the horizontal axis, expanding the inter-word space. Conversely, positive displacement numbers reduce spacing.

In our experimental implementation, a logical bit zero is represented by natural inter-word spacing, while a logical bit one is represented by a subtle negative displacement delta of twenty-five thousandths of an em. At ten-point font size, this corresponds to approximately one-quarter of a point of physical horizontal shift. Because the typographic modification is exceptionally small, visual layout flow and line breaking are preserved without triggering text overflow across standard column gutters.

5. Synchronization Preamble and Error Detection
Inter-word streams are inherently susceptible to synchronization loss if an adversary inserts or deletes content. To guarantee unambiguous frame alignment, our transmission frame incorporates a distinctive thirty-two-bit preamble marker. An automated extractor scans candidate bit sequences, locks onto the preamble, and extracts subsequent payload fields. A thirty-two-bit CRC integrity checksum ensures that corrupted bit sequences resulting from typographic noise are rejected before performing expensive cryptographic operations.

6. Cryptographic Authentication Tag
Rather than truncating the message authentication code to a brief sixty-four-bit digest to minimize carrier payload size, our architecture enforces a full untruncated 256-bit HMAC-SHA256 signature. The canonical message binds the unique document identifier, recipient student credentials, and a fresh 128-bit random nonce. This construction guarantees cryptographic collision resistance and prevents an adversary from forging valid recipient attributions even if the kerning modulation mechanism is completely disclosed.

7. Multi-Column Extraction Robustness
When extracting watermarks from multi-column documents, content-stream extraction parses TJ operators in the exact serialization order emitted by the document layout engine. Because PDF content streams represent text sequentially within each column before transitioning to the adjacent column, the embedded bitstream preserves topological continuity. In contrast, geometric extractors that rely on sorting bounding boxes across the entire page can inadvertently interleave columns if vertical coordinate thresholds are misconfigured.

8. Experimental Observations
Our empirical evaluations demonstrate that multi-column Times-Roman layouts provide ample word boundary density to accommodate full cryptographic frames. Across thirty experimental trials with varied column widths and gutter margins, content-stream parsing achieved one hundred percent recovery with zero bit errors on pristine and lossless re-saved copies.

However, extreme layout modifications such as selective column pruning or re-flowing text into single-column containers disrupt bit sequence continuity. These observations underscore that covert text-level watermarking is fundamentally a vector-representation carrier, effective within digital transmission pipelines but delimited by structural re-authoring.

9. Conclusion
This study demonstrates the viability of high-assurance steganographic watermarking in formal multi-column academic publications. By combining rigorous HMAC-SHA256 signatures with micro-kerning modulation, publishers can achieve unforgeable leak attribution while maintaining full compliance with traditional academic typography.

10. References and Related Literature

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11. Appendix: Cryptographic Security Parameters

All cryptographic primitives in this implementation utilize standard parameters specified by NIST Special Publication 800-107 Revision 1. Message authentication codes employ the HMAC construction with SHA-256 as the underlying cryptographic hash function. Salt nonces are generated using the operating system cryptographically secure pseudorandom number generator (CSPRNG) with one hundred twenty-eight bits of minimum entropy. Key management protocols mandate per-deployment isolated master keys to prevent cross-document correlation attacks.